# LAB 1 — THINKING SHEET

## Learning with Coin Flips

**This sheet is for you. Preparing for the exam starts here.**

**I recommend working in pairs, trying to explain your point of view to each other.**

Running the code is useful, but it is not the goal of this lab. The goal is to practise **predicting, observing, explaining, and experimenting.**

For each investigation: **PREDICT → RUN & OBSERVE → EXPLAIN → EXPERIMENT**

Write short answers. A few words, a sketch, an equation, or a small graph may be enough.

## 1 — Learning with coin flips

### PREDICT — before running

As the number of coin flips increases, what should happen to the estimate of p?

What should happen to its fluctuations? Why?

### RUN & OBSERVE → EXPLAIN

What did you observe, and how is this related to the Law of Large Numbers?

### EXPERIMENT

Change p, the number of flips, or the random seed.

We changed:

We predicted:

We observed:

## 2 — Probability bounds

### PREDICT — before running

As the number of flips increases, what should happen to the distance from the true mean guaranteed by the bounds?

Which bound do you expect to be tighter here? ☐ Chebyshev ☒ Hoeffding ☐ Not sure

### RUN & OBSERVE → EXPLAIN

What did you find? In your own words, what does a probability bound tell us?

### EXPERIMENT

Change δ or p. Prediction: Result:

Explanation:

## 3 — From probability to learning

We now use increasingly rich hypothesis classes and ERM selects one candidate $\theta\square$ from each class.

### BEFORE RUNNING — complete from your understanding

A candidaterepresents:

_____

With squared loss, ERM selects the candidate closest to:

As the hypothesis class becomes richer, predict what will happen:

**Empirical risk** _____

**True risk** _____

**θ□ will tend to move toward** _____

Why?

_____

## 4 — One training set: what actually happened?

Inspect the table and the three plots.

### OBSERVE

As the hypothesis class became richer:

θ□: _____ Empirical risk: _____ True risk: _____

Generalization gap: _____

### EXPLAIN

Why can a richer hypothesis class fit the training data better, but not guarantee a smaller true risk?

_____

_____

Complete the distinction:

ERM is trying to fit _____ while true risk is minimized at _____

Why are these not necessarily the same?

_____

## 5 — Repeat the whole learning experiment

### PREDICT — before running

If we repeat the entire learning experiment many times, what do you expect averaging over repetitions to tell us?

_____

### RUN & OBSERVE → EXPLAIN

What became clearer after averaging over many training sets? Why is the generalization gap itself random?

_____

_____

## END-OF-LAB — CLOSE THE NOTEBOOK

Work individually first.

1. Why can lower empirical risk fail to imply lower true risk?

_____

2. What role does the richness of the hypothesis class play in today's experiment?

_____

3. What is one result from today's lab that you could now explain at the exam without running any code?

_____

Then compare with your partner. One claim you agree on + evidence from today's experiments:

_____